

Changing sea levels

Global sea levels are always rising or falling, often because of climate change. In many places the seabed itself has risen, or the land has sunk beneath the waves. As a result, seabed rocks that contain fossil fish and seashells are now found on land, and some regions that were once land are now shallow seas.

Rock layers revealed in the wall of the Grand Canyon

RISING ROCKS

Many of the rocks that now form much of the land were once soft sediments such as sand and mud, which were laid down on the seabed. They have been turned into rocks—sandstone, shale, and limestone. In places such as the Grand Canyon, you can see many layers of these rocks, all raised high above sea level by the forces that build mountains. Long ago they were under the sea.

ANCIENT OCEANS

We know that many rocks once lay beneath the sea because they contain fossil seashells and fish skeletons. Such fossils have even been found in limestone rocks at the top of Mount Everest, more than 28,200 ft (8,600 m) above sea level. The fossils show that the rocks were formed on the bed of a shallow tropical sea 400 million years ago.



▲ FOSSILIZED AMMONITES

These shells are the remains of sea creatures related to living squid. They are often found in rocks on land.

SINKING SEAS

During the last ice age, which ended about 10,000 years ago, so much rainwater turned into snow and ice that global sea levels fell by about 400 ft (120 m). This exposed vast areas of the continental shelves, which were home to people and land animals such as mammoths. Today, their remains lie under the sea.

► MAMMOTH TOOTH

Fossils of mammoth teeth have been found by fishermen off the Atlantic coast.



◀ DRY LAND

The red dotted lines mark the Atlantic coastlines of ice-age America 22,000 years ago, when mammoths roamed on what is now the continental shelf. The pale blue areas are now shallow seas.